

# Duc Pham

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## EDUCATION

**University of Houston, College of Natural Science and Mathematics**

Houston, TX

*Bachelor of Science in Computer Science | Minor in Management Information Systems*

*Dec 2026 (Expected)*

**GPA: 3.88**

**Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Systems, Data Science

## EXPERIENCE

**Boeing (via LZ Technology)**

Houston, TX

*Software Engineer Intern*

*July 2025 – Present*

- Built a Python-based network automation tool to eliminate manual configuration of network devices, streamlining setup for 20+ devices on board the International Space Station (ISS).
- Optimized core automation logic with robust error handling, retries, and idempotent execution, significantly improving reliability and reducing runtime errors.
- Delivered a production-ready tool enabling ISS operators to configure network devices efficiently and reliably, ensuring operational readiness in space-based workflows.

**ConocoPhillips Learning Center**

Houston, TX

*Academic Support Assistant*

*January 2025 – August 2025*

- Guide students in understanding computer science concepts such as object-oriented programming, data structures, and algorithm fundamentals, ensuring they build a strong foundation in key programming principles.
- Deliver exceptional academic support by tailoring lessons to individual student needs, ensuring their questions and concerns were addressed with care and patience.
- Provide follow-up resources and personalized feedback to reinforce student learning and enhance their understanding of challenging topics.
- **Contributed over 200 hours of academic support** by tutoring and reviewing course materials with students, enhancing their learning outcomes.

## PROJECTS

**Museum Management System** | *Developer* | Node.js, React.js, JavaScript, MySQL, Git January 2025 – May 2025

- Developed a relational database system to manage museum collections, artists, exhibits, and visitor information.
- Implemented SQL queries to manage and retrieve data efficiently, such as fetching all artworks by a specific artist or finding exhibits for specific visitors.
- Collaborated with a team of 4 using Git to develop a full-stack museum database application leveraging React.js for the front-end, JavaScript for the back-end, and MySQL for data storage.

**RPG Project** | *Developer* | C++

October 2023 – December 2023

- Designed and developed a game using a text-based user interface (UI) to facilitate user interaction and decision-making within the game.
- Implemented core aspects of object-oriented programming such as classes, inheritance, encapsulation, abstraction, and polymorphism to efficiently organize and utilize data, enhancing code structure and maintainability.
- Leveraged C++ as the primary programming language for this project, utilizing its features and capabilities to develop and enhance overall performance and functionality.

**Basketball Shot Quality Analysis System** | *Developer* | Python, OpenCV

January 2026 – Present

- Developing a computer vision pipeline to detect and track players and balls across basketball highlight footage.
- Training and evaluating YOLO-based models for small-object tracking, analyzing precision, recall, and mAP to guide optimization.
- Designed custom dataset preprocessing and augmentation pipelines, exploring the impact on model performance (~80% mAP@50).

## TECHNICAL SKILLS, ACTIVITIES, & INTERESTS

**Languages:** Python, C++, JavaScript, HTML/CSS, React, SQL, R

**Developer Tools:** Git, VS Code, Visual Studio, Shell

**Certifications:** Microsoft Introduction to Programming using Python (August 2021 – May 2022)